

Resonant ckt

Resonance is a condition in RLC ckt in which capacitance & inductive reactances are equal in magnitude, thereby resulting in purely resistive ckt.

Following conditions are required for resonance

① $X_L = X_C$

② $Z = R$ overall resistance is $= R$

③ power factor $\cos \phi = 1$ unity

④ phase shift is equal to zero.

Application: ① In communication.

② Oscillators

④ Filters

③ Tuning ckt

⑤ Medical & synthetic equip. meas.

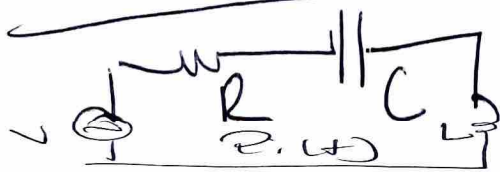
Resonance occurs only in RLC ckt.

There are 2 types

① Series resonant ckt

② Parallel " "

Series resonant ckt: Resonance is a condition in which Z attains minimum value & I attains maximum value.



ckt becomes maximum.

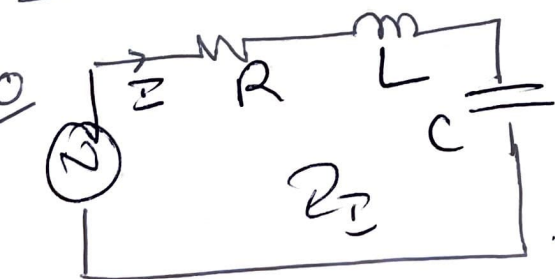
Parallel resonant ckt :

Resonance is a condition in which the C in the circuit will attain minimum value & impedance in the circuit becomes maximum.

Quality factor in series RLC ckt

Quality factor measures sharpness of the resonance. Q of the RLC ckt high indicates energy storing capacity is high. It is also the ratio of stored energy to dissipated energy.

Series RLC ckt :



$$Z = R + j\omega L + \frac{1}{j\omega C}$$

$$Z = R + j\omega L - j\omega C$$

$$v_s(t) = V_s \sin \omega t$$

$$Z = R + j[\omega L - \omega C]$$

WKT $\omega L = 2\pi fL$

$$\omega C = \frac{1}{2\pi fC}$$

$\omega = 2\pi f$ SO $\omega L = \omega L$

$\omega C = \frac{1}{\omega C}$

At resonance $\omega L = \omega C$

ie $\omega L = \frac{1}{\omega C}$

$$\omega^2 = \frac{1}{LC}$$

$\omega = \frac{1}{\sqrt{LC}}$ rad/sec
is called resonant freq.

We vary the frequency ω & note ω when

$\omega = \omega_r = \omega_0 = \frac{1}{\sqrt{LC}}$ rad/sec we can say

resonance occurs.

Also we can change the value of L & C to get resonance in the ckt.

At resonance

The impedance $Z = R + j(\omega L - \frac{1}{\omega C})$

The magnitude $|Z| = \sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}$

$$Z = \frac{V}{|Z|} = \frac{V}{R}$$

$\therefore Z = R$

At resonance $|Z| = R$

Let The v_g across the inductor

$$V_L = I X_L$$

$$= \frac{V}{R} \omega L$$

$$= \frac{\omega L}{R} V$$

$$V_L = QV$$

$$Q = \frac{\omega L}{R}$$

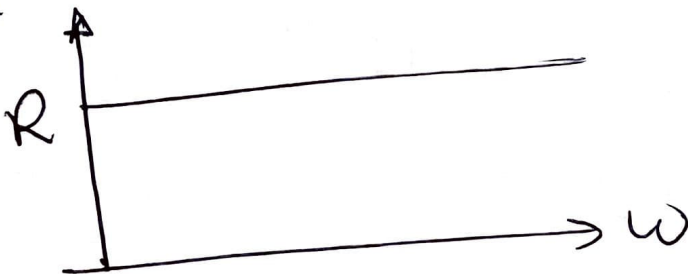
where $V =$ supply v_g
 $V_L =$ v_g across inductor.

$$V_C = I X_C = \frac{V}{R \omega C} = \frac{V}{R \omega C} = QV$$

$$V_C = QV$$

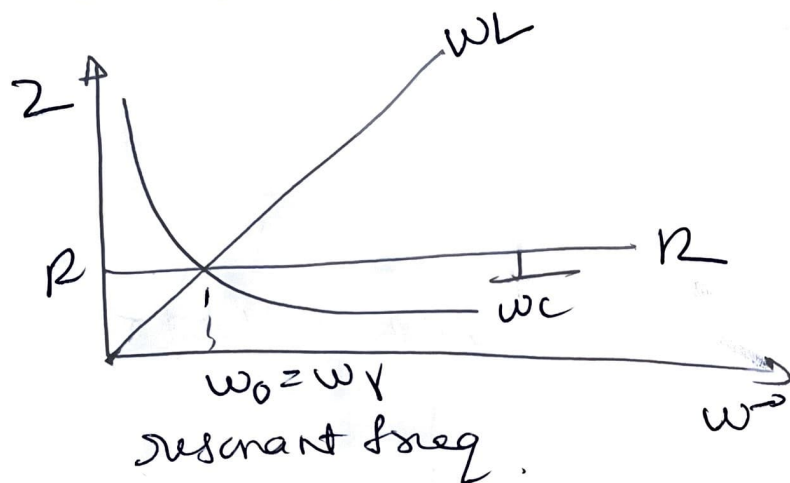
$\therefore Q$ is also called magnitude of voltage.

Graphs of RLC ckt



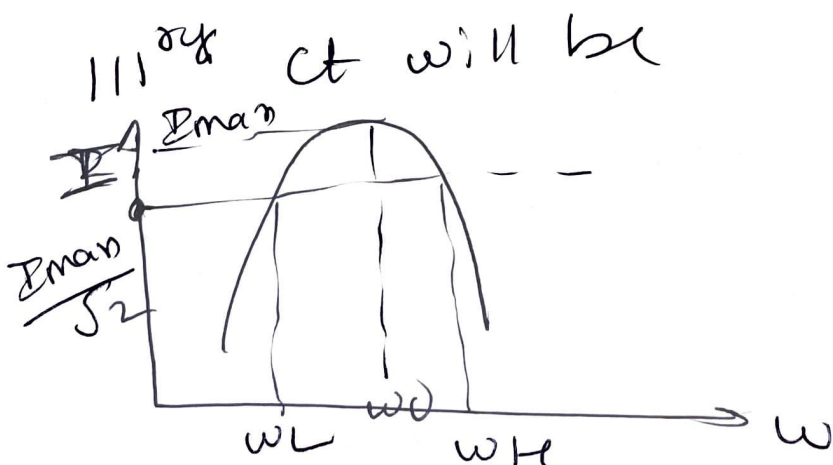
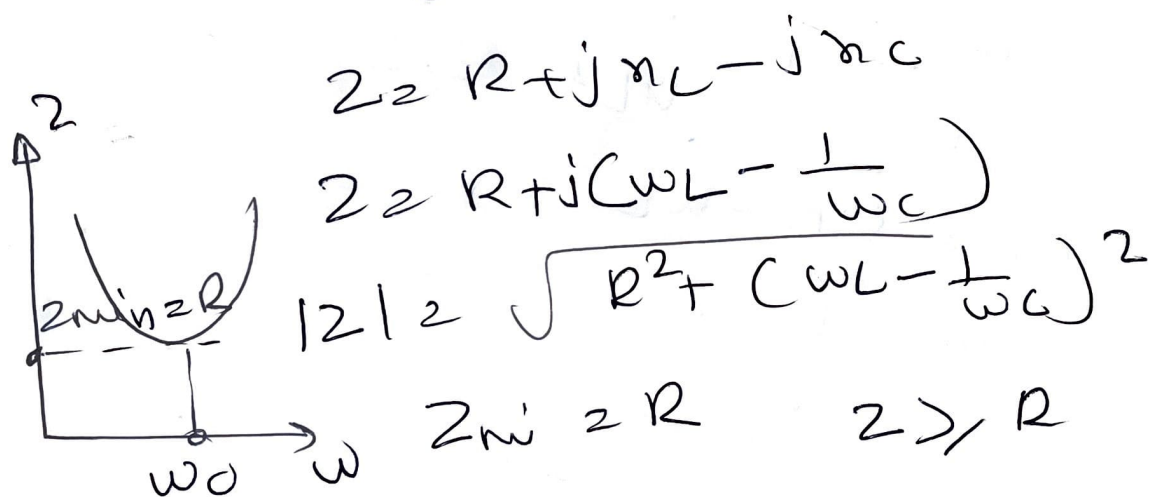


Combine all the above 3 graphs we can write as



Here the effect of inductor is increasing & effect of capacitor is decreasing $\therefore$ both the effect

get cancelled. & only R remains in the circuit so the graph will be



$w_H \rightarrow$ higher cut off freq
 $w_L \rightarrow$ lower cut off freq.



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Supplementary Examination Summer term July/ Aug 2024

Attendance Report

FORM-B

$$|I| = \frac{V}{Z} = \frac{V}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}}$$

$$I = \frac{I_{max}}{\sqrt{2}} = \frac{V}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}}$$

$$\omega_H = \frac{R}{2L} + \sqrt{\frac{R^2}{4L^2} + \frac{1}{LC}}$$

$$\omega_L = -\frac{R}{2L} + \sqrt{\frac{R^2}{4L^2} + \frac{1}{LC}}$$

$$BW = \omega_H - \omega_L$$

$$BW = \frac{R}{L} \text{ rad/sec}$$

$$Q \cdot BW = \omega_0 = \sqrt{\omega_H \omega_L}$$

Quality factor Q in Series RLC Ckt

Quality factor measures the sharpness of

①

the resonance

$$\omega_0 = \omega_L = \omega_C$$

$$Q = \frac{\omega_0 L}{R}$$

$$Q = \frac{1}{\sqrt{LC}} \times \frac{L}{R} = \frac{1}{R} \times \frac{L}{\sqrt{LC}}$$

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$$

② Q of RLC ckt. high indicates energy storing capacity is high. Q is the ratio of stored energy to dissipated energy

$$Q = \frac{2\pi \times \text{Max. energy stored/cycle}}{\text{Energy dissipated / cycle.}}$$

Consider the series RL ckt & the avg V is applied to the ckt. The maximum energy stored/cycle is

$$W_L = \frac{1}{2} L I_m^2$$

The avg power dissipated is given by $I_m = \text{max ct}$

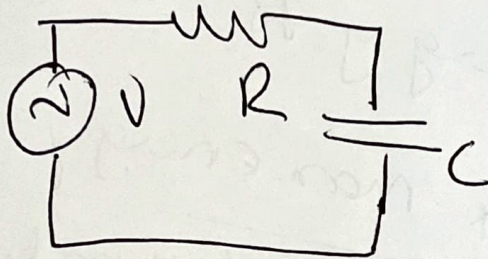
$$P = \left(\frac{I_m}{\sqrt{2}} \right)^2 R = \frac{1}{2} I_m^2 R$$

The total energy dissipated is given by $= \frac{P}{f}$

$$Q = \frac{2\pi \frac{1}{2} L I_m^2}{\frac{1}{2} \frac{I_m^2 R}{f}} = \frac{2\pi f L}{R} = \frac{\omega L}{R}$$

$$= \frac{1}{2} \frac{I_m^2 R}{f}$$

Now let us consider the the ckt is now
in series with capacitor



$$W_C = \frac{1}{2} C V_m^2$$

$V_m \rightarrow$ max peak

$$= \frac{1}{2} C \left[\frac{I_m}{\omega C} \right]^2$$

applied voltage

$$= \frac{1}{2} \times \frac{I_m^2}{\omega^2 C}$$

$$W_C = \frac{1}{2} \frac{I_m^2}{\omega^2 C}$$

avg power dissipated

$$P_D = \left(\frac{I_m}{\sqrt{2}} \right)^2 R = \frac{1}{2} I_m^2 R$$

The total energy dissipated/cycle is given by

$$= \frac{P_D}{f}$$

$$= \frac{\frac{1}{2} I_m^2 R}{f}$$

$f \rightarrow$ operating freq

$$Q = 2\pi \times \frac{\frac{1}{2} I_m^2}{\omega^2 C}$$

$$= 2\pi \times \frac{1}{\omega^2 C} \times \frac{f}{R}$$

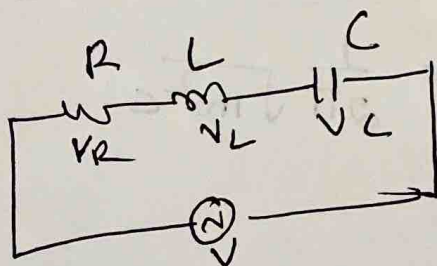
$$\frac{\frac{1}{2} I_m^2 R}{f}$$

$$= 2\pi \times \frac{1}{(2\pi f)} \times \frac{f}{CR}$$

$$Q = \frac{1}{\omega CR}$$

Q factor depends on both R & C values.

Series resonance:



The impedance RLC is given by

$$Z = \frac{V}{I}$$

$$Z = R + j\omega L - j\omega C$$

$$= R + j(\omega L - \omega C)$$

$$= R + j\left[\omega L - \frac{1}{\omega C}\right]$$

$\omega \rightarrow$ freq in rad/sec.

The definition of resonance is reactive part in impedance RLC circuit is zero. $\omega_0 \rightarrow$ resonant freq.

[for resonance V_R is std bc in phase & Resonance applicable only to AC ckt]



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FORM-B

At resonance

$$\omega_0 L - \frac{1}{\omega_0 C} = 0$$

$$\omega_0^2 = \frac{1}{LC}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/sec}$$

$$\omega_0 = 2\pi f_0$$

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \text{ Hz //$$

Problem :- A series RLC ckt driven with a sinusoidal ac voltage source, determine value of C required to achieve resonance in ckt at 5 kHz if the value of $R = 2 \Omega$ & $L = 1 \text{ mH}$.

Soln :- $R = 2 \Omega$ $L = 1 \text{ mH}$ $C = ?$ $f_0 = 5 \text{ kHz}$

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = 5 \text{ K} = \frac{1}{2\pi\sqrt{1 \times 10^{-3} C}}$$

Squaring

$$(5 \text{ K})^2 = \frac{1}{(2\pi)^2 (1 \times 10^{-3})^2 C}$$

$$C = 1.0132 \mu\text{F}$$

A coil of 5mH inductance & 10Ω resistance is connected in series with $5\mu\text{F}$ capacitor. determine the freq at which ckt resonates
 at which V & I are in phase

Soln $R = 10$ $L = 5\text{mH} = L = 5\mu\text{F}$ $f = ?$

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{5\text{m} \times 5\mu\text{F}}}$$

$$= 1.0065\text{kHz}$$

2) In a series RLC ckt driven with a sinusoidal ac voltage source, determine value of C required to achieve resonance in ckt at 5kHz if value of resistance & inductance are 2Ω & 1mH respectively

Soln $C = ?$
 $f = 5\text{kHz}$
 $R = 2$
 $L = 1\text{m}$

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

squaring

$$f^2 = \frac{1}{4\pi^2 LC}$$

$$C = \frac{1}{4\pi^2 L f^2}$$

$$C = 1.013\mu\text{F}$$

properties of resonant ckt

- Resonant ckt is basically a freq selective ckt
- The behavior of the ckt changes in variation with operating frequency.
- if $x_L = \omega L = 2\pi f L$

The graph can be represented as



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FORM-B

D A coil of inductance 0.1 H & resistance of 10Ω is connected in series with capacitor of $0.1 \mu\text{F}$. Find freq of resonance of the circuit. Also find the quality factor of the circuit at resonance.

Soln $L = 0.1$ $R = 10$ $C = 0.1 \mu\text{F}$

$$① f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{0.1 \times 0.1 \times 10^{-6}}} = 1.59 \text{ KH.}$$

$$② Q = \frac{\omega_0 L}{R} = \frac{2\pi f_0 L}{R} = \frac{2\pi \times 1.59 \times 0.1}{10}$$

$$Q = 99.99 \approx 100$$

we can also calculate $Q = \frac{1}{R} \sqrt{\frac{L}{C}}$

$$= \frac{1}{10} \sqrt{\frac{0.1}{0.1 \times 10^{-6}}} = 100$$

Band width of series RLC resonant ckt

BW of the resonant ckt is band of freq over which the power in the ckt is half of the max. value.

At resonance the $I_0 = \frac{V}{R}$. It is $\neq$ imp

The resonant freq the power

$$P_0 = P_{\text{max}} = I_0^2 R$$

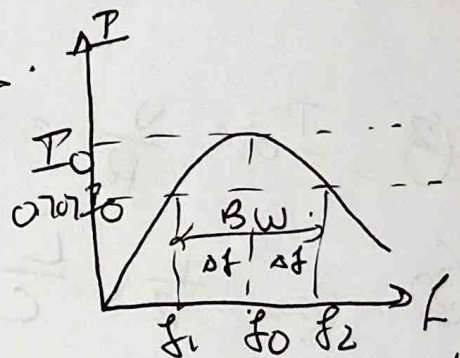
Now half of the max. power is

$$P' = \frac{P_0}{2} = \frac{I_0^2 R}{2} = \left(\frac{I_0}{\sqrt{2}} \right)^2 R$$

$$= \frac{I_0^2 R}{2}$$

$$P' = \frac{I_0^2 R}{2}$$

$$\boxed{BW = f_2 - f_1}$$



$f_1 \rightarrow$ Lower half power freq
 $f_2 \rightarrow$ upper " " "

At resonance the max. power P' in the ckt is

0.707 times the resonant I_0 .

Selectivity:- Resonant ckt are also called freq selective ckt. Selectivity is an ability of ckt to discriminate or distinguish b/w desired & undesired freq.

$$\text{also Selectivity} = \frac{\text{resonant freq}}{BW} = \frac{f_0}{f_2 - f_1}$$

A series RLC circuit consist of resistance $1k\Omega$ & an inductance of $100mH$ in series with capacitor $10pF$. If $100V$ is applied as i/p across the combination determine

① The resonant freq.

② Max-Ct in the Ckt

③ Q factor

④ ~~band~~ half power frequencies.

Sol $R = 1k\Omega$ $L = 100m$ $C = 10pF$ $V = 100$

① Resonant freq $f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{100m \times 10p}}$
 $= \underline{\underline{159.15\text{KHz}}}$

② $I_0 = \frac{V}{R} = \frac{100}{1k} = 0.1A$

③ $Q = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{1}{1k} \sqrt{\frac{100m}{10 \times 10^{-12}}} = 100$

④ $BW = f_2 - f_1 = \frac{R}{2\pi L} = \frac{1k}{2\pi \times 100m} =$
 $= 2\Delta f$

$2\Delta f = \frac{R}{2\pi L}$

$\Delta f = \frac{R}{4\pi L} = \frac{1k}{4\pi \times 100m}$

$= 795.77\text{Hz}$

Lower half power freq $f_1 = f_0 - \Delta f$
 Upper " " " " $f_2 = f_0 + \Delta f$

~~$2\Delta f = f_2 - f_1$~~

~~$2\Delta f =$~~

$f_1 = 158.35\text{KHz}$

$f_2 = 159.95\text{KHz}$

capacitor

Derivation to get Half power frequency

Let $\omega_H \rightarrow$ Higher cut off freq in rad/sec
 $\omega_L \rightarrow$ Lower " " in rad/sec

Why f_H & f_L are higher & lower cut off freq.

WKT $P = \frac{1}{2} P_{max}$

$$P = \frac{P_{max}}{\sqrt{2}}$$

$$P = \frac{V}{|Z|} \cdot \frac{V}{|Z|} = \frac{P_{max} R}{|Z| \sqrt{2}} = \frac{P R}{\sqrt{2} R} \quad \begin{matrix} V = IR \\ Z = R \end{matrix}$$

$$= \frac{V}{|Z|} = \frac{V}{\sqrt{2} R}$$

$$|Z| = \sqrt{2} R$$

~~$$Z = R + j\omega L - \frac{j}{\omega C}$$~~

$$Z = R + j\omega L - \frac{j}{\omega C}$$

$$|Z| = \sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}$$

$$\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2} = \sqrt{2} R$$

Squaring Both the side

$$R^2 + (\omega L - \frac{1}{\omega C})^2 = 2R^2$$

$$(\omega L - \frac{1}{\omega C})^2 = R^2$$

$$\omega L - \frac{1}{\omega C} = \pm R$$



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FORM-B

Note: All the entries in the Form B except Candidate Signature should be strictly done by the Invigilator only after

verifying the details from the College ID Card and Hall Ticket of the Candidate.

$$\omega L - \frac{1}{\omega C} + R = 0$$

x'ly by ω

~~$$\omega^2 L - \frac{1}{C} + R\omega = 0$$~~

$$\omega^2 L - \frac{1}{C} + R\omega = 0$$

$$\omega^2 L + R\omega - \frac{1}{C} = 0$$

Roots factor $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$\omega_1 = \frac{-R \pm \sqrt{R^2 + \frac{4L}{C}}}{2L}$$

$$\omega_2 = \frac{R \pm \sqrt{R^2 + \frac{4L}{C}}}{2L}$$

$$\omega_1 = -\frac{R}{2L} + \sqrt{\frac{R^2}{4L^2} + \frac{1}{LC}}$$

~~$$\omega_2 = \frac{R}{2L} + \sqrt{\frac{R^2}{4L^2} + \frac{1}{LC}}$$~~

$$BW = f_2 - f_1 = \frac{2R}{2L} = \frac{R}{L}$$

$$\omega_2 = \frac{R}{2L} + \sqrt{\frac{R^2}{4L^2} + \frac{1}{LC}}$$

$$BW = \frac{R}{2\pi L}$$

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$$BW = \omega_2 - \omega_1 = \frac{R}{L}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

At half power frequencies the resonance is exactly in the middle.

$$\text{i.e. } \omega_0 = \frac{\omega_1 + \omega_2}{2} \quad \text{or} \quad \frac{\omega_L + \omega_H}{2}$$

$$\omega_1 + \omega_2 = 2\omega_0 \quad \text{--- (1)}$$

$$\omega_2 - \omega_1 = BW = \frac{R}{L} \quad \text{--- (2)}$$

Adding eqn (1) & (2)

$$\omega_1 + \omega_2 + \omega_2 - \omega_1 = 2\omega_0 + \frac{R}{L}$$

$$2\omega_2 = 2\omega_0 + \frac{R}{L}$$

$\div$ by 2

$$\omega_2 = \omega_0 + \frac{R}{2L} \quad \text{--- (3)}$$

subtract both the eqn

$$\omega_1 + \omega_2 - \omega_2 + \omega_1 = 2\omega_0 - \frac{R}{L}$$

$$\frac{2\omega_1}{2} = \frac{2\omega_0}{2} - \frac{R}{2L}$$

$$\omega_1 = \omega_0 - \frac{R}{2L} \quad \text{--- } \textcircled{\omega}$$

$$\text{ie } \omega = 2\pi f$$

$$\omega \times T \quad f = \frac{\omega}{2\pi}$$

$$f_1 = \frac{\omega_1}{2\pi} \quad \& \quad f_1 = \frac{\omega_0 - \frac{R}{2L}}{2\pi}$$

$$f_1 = \frac{\omega_0}{2\pi} - \frac{R}{2L \times 2\pi} = \frac{\omega_0}{2\pi} - \frac{R}{4\pi L}$$

$$f_1 = \frac{\omega_0}{2\pi} - \frac{R}{4\pi L}$$

$$\omega_0 = 2\pi f_0$$

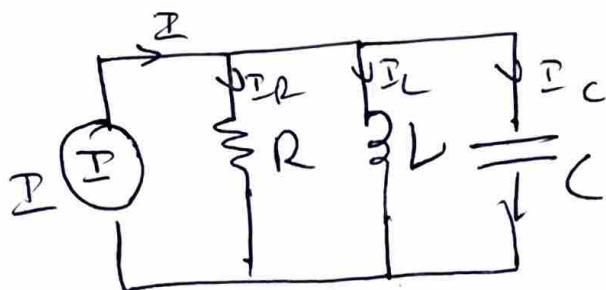
$$f_1 = f_0 - \frac{R}{4\pi L} = f_0 - \frac{BW}{2}$$

$$\text{III} \quad f_2 = f_0 + \frac{R}{4\pi L} = f_0 + \frac{BW}{2}$$

$$BW \quad f_2 - f_1 = f_0 + \frac{R}{4\pi L} - f_0 + \frac{R}{4\pi L}$$

$$BW = \frac{2R}{4\pi L} = \frac{R}{2\pi L}$$

Parallel Resonant Ckt



WKT $I = I_R + jI_C - jI_L$ $Y = \frac{1}{R}$

$$I = I_R + j(I_C - I_L)$$

$$Y_{eq} = \frac{1}{R} + \frac{1}{j\omega L} + \frac{1}{1/j\omega C}$$

$$= \frac{1}{R} + j\omega C + \frac{1}{j\omega L}$$

$$Y_{eq} = \frac{1}{R + j\omega L} + j\omega C$$

or

$$Y_{eq} = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)$$

$$Y_{eq} = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right) \quad \text{--- (1)}$$

$$|Y_{eq}| = \sqrt{\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2}$$

equating imaginary part to zero in eqn (1)

$$\omega C = \frac{1}{\omega L}$$

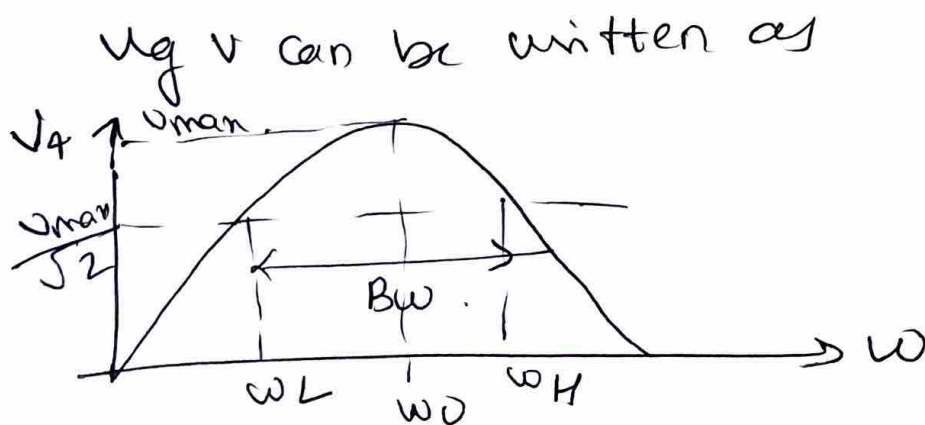
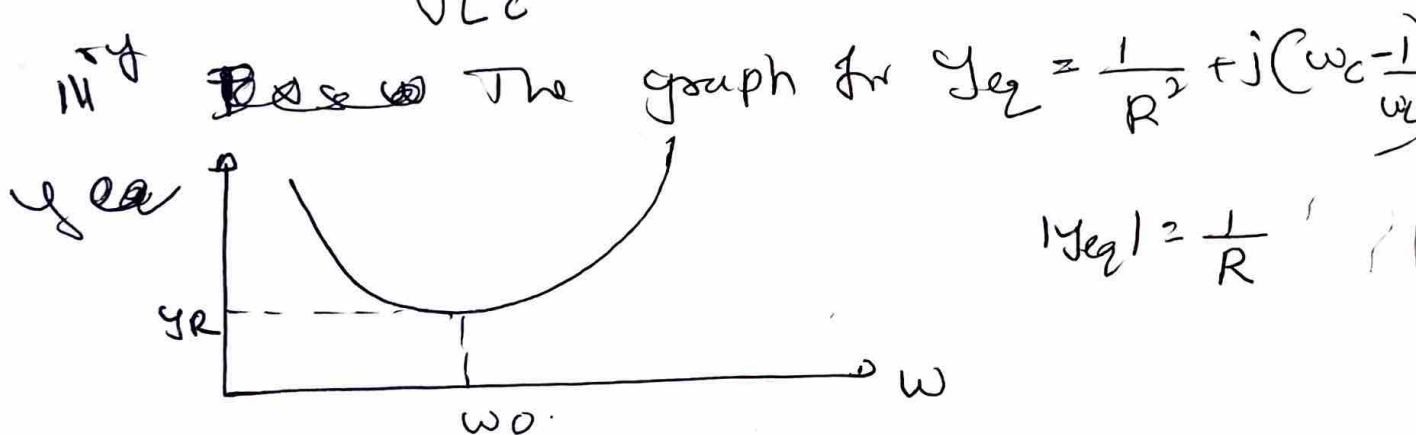
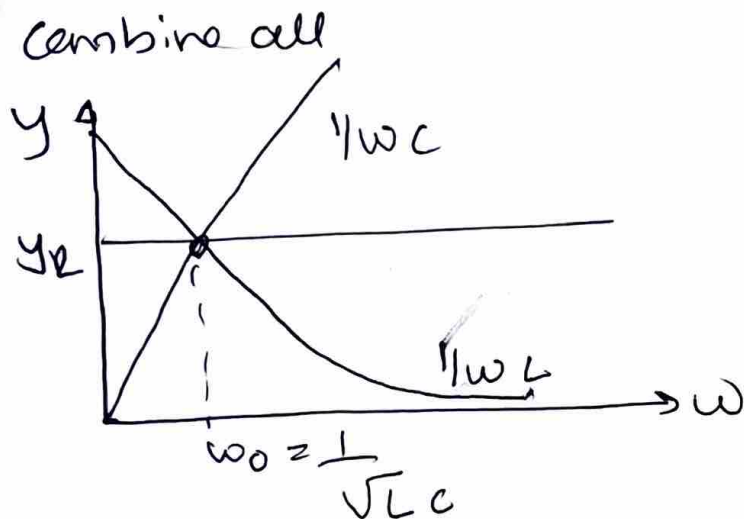
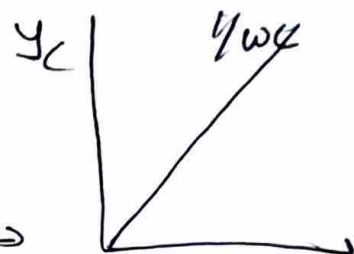
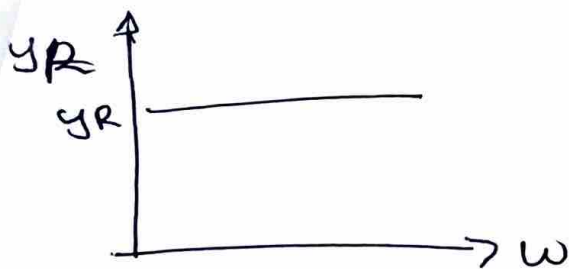
$$\omega^2 = \frac{1}{LC}$$

$$\omega = \frac{1}{\sqrt{LC}}$$

rad/sec.

Same as in series ckt the admittance

graph is as below



$Z = \frac{1}{Y}$

$V \propto \frac{1}{Y}$

$V = I Z$

$V = \frac{I}{Y}$

$V = \frac{I}{\sqrt{\frac{1}{R^2} + (\omega C - \frac{1}{\omega L})^2}}$

A series RLC ckt with $R = 10\Omega$, $L = 1\text{mH}$ & $C = 1000\text{pF}$ is connected across sinusoidal source of 20V with variable freq.

a) compute the resonant frequency of the ckt

b) Find Q at resonant freq.

c) Determine the half power frequencies

Ans $R = 10\Omega$ $L = 1\text{mH}$ $C = 1000\text{pF}$
 $= 159155$

a) $f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{1\text{m} \times 1000 \times 10^{-12}}}$

b) $Q = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{1}{10} \sqrt{\frac{1\text{m}}{1000 \times 10^{-12}}} = 100$

c) $f_1 = f_0 - \frac{R}{4\pi L} = 159155 - \frac{10}{4\pi \times 1\text{m}} = 158359$

$f_2 = f_0 + \frac{R}{4\pi L} = 159155 + \frac{10}{4\pi \times 1\text{m}} = 159951$

> A series RLC ckt with $Q = 250$ is resonant at 1.5MHz . Find the frequencies at half power points also find the BW.

① $Q = 250$

$f_0 = 1.5\text{MHz}$

$f_1 = ?$ $f_2 = ?$

① $Q = \frac{f_0}{\text{BW}}$

$\text{BW} = \frac{f_0}{Q} = \frac{1.5\text{M}}{250} = 6\text{kHz}$

$= 1.5\text{M} - \frac{6\text{K}}{2} =$

$f_1 = f_0 - \frac{\text{BW}}{2}$

$= 1.5\text{M} + \frac{6\text{K}}{2} =$

$f_2 = f_0 + \frac{\text{BW}}{2}$



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FORM-B

- ① A Coil of resistance 2Ω and inductance $0.01H$ is connected in series with a capacitor C .
If the max. ~~it occurs at~~ $25Hz$ find C
resonant freq f_0

Sol

① $L = 0.01H$

$f_{max} = ?$

$f_0 = 25Hz$

$C = ?$

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$25 = \frac{1}{2\pi\sqrt{0.01 \times C}}$$

$$C = 1053\mu F$$

- ② what is the resonant freq and BW of a Series RLC ckt given $R = 55\Omega$ & $L = 40mH$
 $C = 1\mu F$

Soln

$R = 55\Omega$

$L = 40mH$

$C = 1\mu F$

$f_0 = ?$

$BW = ?$

$= 795.8Hz$

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{40m \times 1\mu}}$$

$BW = \frac{\omega L}{R}$

$$Q = \frac{\omega L}{R} = \frac{2\pi f L}{R} = \frac{2\pi \times 795.8 \times 40m}{55\Omega}$$

$Q = 90.865 \times 10^{-3}$

$Q = \frac{f_0}{BW}$

$BW = \frac{795.8}{90.86} = 8.75$
 $BW = 20$

A coil of resistance $40\ \Omega$ & inductance 0.75 mH forms part of series circuit for which the resonant frequency is 55 kHz . If the supply is 250 V , 50 kHz find the value of C , impedance Z & current I

Soln $R = 40\ \Omega$ $L = 0.75\text{ mH}$ $f_0 = 55\text{ kHz}$ $V = 250$
 $C = ?$

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = 55 = \frac{1}{2\pi\sqrt{0.75 \times 10^{-3} C}}$$

$$C = 11.16\ \mu\text{F}$$

$$Z = R + j(\omega L - \frac{1}{\omega C}) = R + j(\omega L - \frac{1}{\omega C})$$

$$= R + j(2\pi fL - \frac{1}{2\pi fC})$$

$$|Z| = \sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}$$

$$= \sqrt{40^2 + (235.6 + 285.6)^2}$$

$$= \underline{\underline{63.72}}$$

$$\omega L = 2\pi fL$$

$$\omega L = \frac{1}{2\pi fC}$$

$$\omega L = 235.6$$

$$\omega C = 285.2$$

$$I = \frac{V}{|Z|} = \frac{250}{63.72} = 3.92$$

7. A series RLC circuit has $R = 200\ \Omega$ $L = 20\text{ mH}$ $C = 0.5\ \mu\text{F}$. Calculate the max C in the circuit & the avg across L & C for i/p v of 200 V

Soln



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Supplementary Examination Summer term July/ Aug 2024
Attendance Report

FORM-B

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{20m \times 0.5\mu}} = 1592 \text{ Hz}$$

$$I_{max} = \frac{V}{R} = \frac{200}{200} = 1 \text{ A}$$

$$V_L = ? \quad V_C = ?$$

$$V_L = I \times L = 1 \times 2\pi f_0 \times 20m = 2\pi \times 1592 \times 20m = 200 \text{ V}$$

$$V_C = I \times C = 1 \times \frac{1}{2\pi f_0 C} = \frac{1}{2\pi \times 1592 \times 0.5\mu} = 200 \text{ V}$$

$L = 20mH$

$C = 20.01\mu F$

$V = 200$